5 Find the particular solution of the differential equation

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 4\cos x,$$

given that, when $x = 0$, $y = -4$ and $\frac{dy}{dx} = 3$. [11]

7 **(a)** Show that an appropriate integrating factor for

$$\sqrt{x^2-1}\,\frac{\mathrm{d}y}{\mathrm{d}x}+y=x^2-x\sqrt{x^2-1}$$

is $x+\sqrt{x^2-1}$. [4]

(b) Hence find the solution of the differential equation

$$\sqrt{x^2-1}\,\frac{\mathrm{d}y}{\mathrm{d}x}+y=x^2-x\sqrt{x^2-1}$$

for which $y = 1$ when $x = \frac{5}{4}$. Give your answer in the form $y = \mathrm{f}(x)$. [7]